

Adroit Technologies

Protocol Driver

Name	Endress & Hauser Promag 800 Email
DLL	EHPROMAG



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1. Overview

This driver is designed to provide an interface for Promag 800 flow meters.

The data is sent in the form of an email containing a csv attachment.

Once the email is received it is decoded to expose the values contained in the attachment and body of the mail.

Inbox management is provided through a BACKUPBOX mailbox and an UNKNOWN/TODO mailbox, where email messages can be moved as appropriate.

Messages older than a configured amount of days may be set for auto deletion.

A separate Adroit device has to be created for each flow meter with the station ID being set equal to the serial number of the flow meter as per the CSV file that get emailed.

There is a resource ID allocated automatically on the device configuration, where new devices will attempt to find a compatible configuration (those that share an inbox for example). Any others will create a new resource. Selecting a resource in the dialog will load the current configuration for that box. If the device is saved (on OK), then any changes will be saved with an appropriate warning. Resource ID can be deleted with appropriate warnings if it might affect other devices.

Offline devices and unknown (unconfigured) devices will have their emails moved to the UNKNOWN box if the MOVE radio is applied. Else if DELETE was selected, then unconfigured emails will be deleted, and offline device emails will be left in the main INBOX.

2. Revisions

2.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	General edits, formatting and content additions.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

2.2. DLL Revision

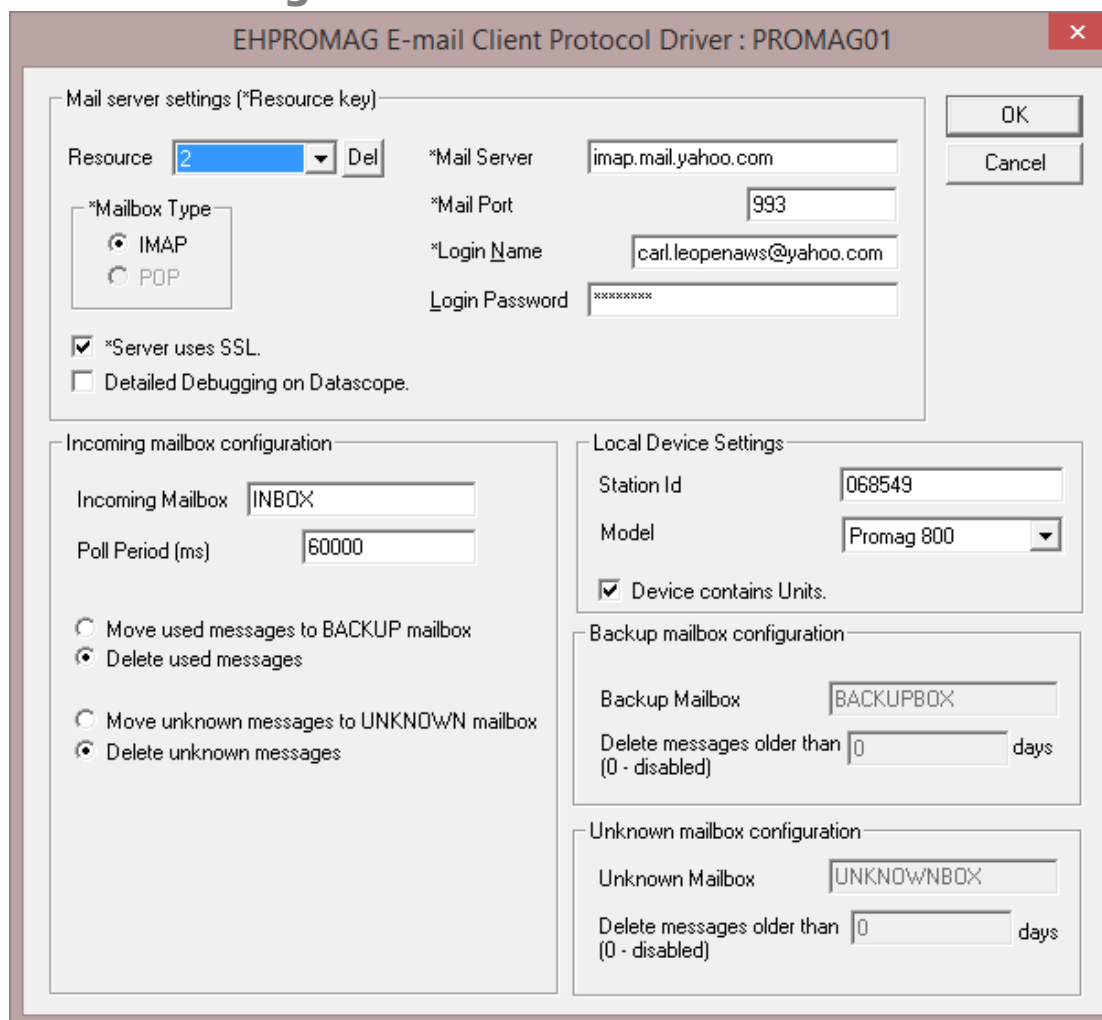
Rev.	Date	Changes
	16-Apr-2015	First Released Document.

3. Wiring

Not applicable.

4. Device Configuration

4.1. Device Configuration



4.1.1. Mail Server Settings (*Resource key)



PLEASE NOTE

The parameters identified by the * define a resource common to all devices that share those parameters, any change to any other parameter in this group will be changed for all devices using this mailbox.

***Mailbox Type** – Choose IMAP or POP depending on your server type (**POP is no longer supported**).

***Server uses SSL** – (Default = ON) Some POP and IMAP servers require SLL mode. IMAP servers listen on port 143 typically, while **993** is used for SSL/TLS.
POP servers listen on port 110 typically, while **995** is used for SSL/TLS (**no longer supported**).

***Mail Server** – The name of the mail server.

***Mail Port** – 993 for IMAP (**143 is now no longer used**), 110 for POP (**no longer supported**).

***Login Name** – Username on the server.



PLEASE NOTE

The remaining parameters can be considered as sub-parameters (these will not result in a new resource being generated if different from any other resource) and if a change is requested on these the whole resource will be updated, and thus affect all devices using that resource.

Login Password – User password.

Detailed Debugging to Datascope – Additional processing information will be sent to the datascope.

4.1.2. Incoming Mailbox Configuration

Incoming Mailbox - Incoming account inbox name. This is where all new emails should arrive first.

Poll Period (ms) – Mailbox will be checked for new mail at this interval.

Move used messages to BACKUP mailbox – If a mail is parsed, accepted and considered complete, checking this radio button will tell the driver to move that mail to the used or BACKUP mailbox, where further processing could take place, such as remain there for x amount of days before being discarded (see further down).

Delete used messages – If a mail is parsed and accepted and considered complete, checking this radio button will tell the driver to delete the mail immediately.

Move unknown messages to UNKNOWN mailbox – If a mail is parsed and found to be junk, or for an offline device and NOT considered complete, checking this radio button will tell the driver to move that mail to the TODO or UNKNOWN mailbox, where further processing could take place such as remain there for x amount of days before being discarded (see further down).

Delete unknown messages – If a mail is parsed and found to be junk, or for an offline device and is not considered complete, checking this radio button will tell the driver to delete the mail immediately.

4.1.3. Local Device Settings



PLEASE NOTE

These setting are stored per device and will not affect other devices using the same resource ID.

Station ID – This is the identifier code sent in the subject line of the email. For example, data from ML250 29902, the ID is then 29902.

Model – Select the Model of the Promag 800 flowmeter. Currently only Promag 800 is available.

Device contains units – The Promag 800 can report data with unit fields.

4.1.4. Backup mailbox configuration

Backup Mailbox – Backup name – this is a sub or secondary mailbox where completed emails can be stored for a period of time.

Delete messages older than x days – Any email in this box older than this number of days will be deleted during driver maintenance tasks.

4.1.5. Unknown Mailbox Configuration

Unknown Mailbox – Unknown name – this is a sub or secondary mailbox where incomplete emails can be stored for a period of time. Junk/spam and or unconfigured or offline devices will also go here.

Delete messages older than x days – Any email in this box older than this number of days will be deleted during driver maintenance tasks.

5. Supported Addresses

A CSV row constitutes 1 record:

1,2010/02/03,00:00:00,<Unit 1>,<Value1>,<Unit 2>,<Value2>...<Unit 21>,<Value21>

Data	Result Representation	Adroit Integer	Adroit Real	Adroit String
Unit 1	Unit for Positive Total			U1
Value 1	Value of Positive Total	V1	V1	
Unit 2	Unit of Negative Total			U2
Value 2	Value of Negative Total	V2	V2	
Unit 3	Unit of Flow Rate			U3
Value 3	Value of Flow Rate	V3	V3	
Unit 4	Unit of Flow Rate %			U4
Value 4	Value of Flow Rate %	V4	V4	
Unit 5	Unit of SPARE			U5
Value 5	Value of SPARE	V5	V5	
Unit 6	Unit of SPARE			U6
Value 6	Value of SPARE	V6	V6	
Unit 7	Unit of Positive Partial Total			U7
Value 7	Value of Positive Partial Total	V7	V7	
Unit 8	Unit of Negative Partial Total			U8
Value 8	Value of Negative Partial Total	V8	V8	
Unit 9	Unit of Total Net Value			U9
Value 9	Value of Total Net Value	V9	V9	
Unit 10	Unit of Partial Net Value			U10
Value 10	Value of Partial net Value	V10	V10	
Unit 11	Unit of SPARE			U11
Value 11	Value of SPARE	V11	V11	
Unit 12	Unit of SPARE			U12
Value 12	Value of SPARE	V12	V12	
Unit 13	Unit of SPARE			U13
Value 13	Value of SPARE	V13	V13	
Unit 13	Unit of SPARE			U14
Value 14	Value of SPARE	V14	V14	
Unit 15	Unit of SPARE			U15
Value 15	Value of SPARE	V15	V15	
Unit 16	Unit of Cycles Per Hour			U16
Value 16	Value of Cycles Per Hour	V16	V16	
Unit 17	Unit of Battery 1 Status			U17
Value 17	Value of Battery 1 Status	V17	V17	
Unit 18	Unit of Battery 2 Status			U18
Value 18	Value of Battery 2 Status	V18	V18	
Unit 19	Unit of Battery 3 Status			U19
Value 19	Value of Battery 3 Status	V19	V19	
Unit 20	Unit of Antenna Signal			U20
Value 20	Value of Antenna Signal	V20	V20	
Unit 21	Unit of Board Temperature			U21
Value 21	Value of Board Temperature	V21	V21	

6. Protocol Definition and Considerations

6.1. Application example

1. You have a Promag 800 flow meter from which you need to extract the following data:
 - a. The positive volume counter
 - b. The negative volume counter
 - c. The Flow rate
2. You are also required to extract the units of these values from the csv file.
3. In addition to this you need to determine the signal strength of the antenna.

In order to accomplish this, you will need to create 4 Analog Agents and 4 String Agents.

You will scan the value of the positive volume counter (analog) from the "V1" address of the device.

You will scan the unit of the positive volume counter (string) from the "U1" address of the device.

You will scan the value of the negative volume counter (analog) from the "V2" address of the device.

You will scan the unit of the negative volume counter (string) from the "U2" address of the device.

You will scan the value of the flow rate (analog) from the "V3" address of the device.

You will scan the unit of the flow rate (string) from the "U3" address of the device.

You will scan the value of the antenna signal strength from the "V20" address of the device.

You will scan the unit of the antenna signal strength from the "U20" address of the device.

7. General Notes and Troubleshooting

7.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

7.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7.3. Fault finding

1. Ensure the connection settings to your mailbox is correct.
2. Ensure the serial number of the Promag 800 flow meter matches the Station ID as set up in the driver configuration.
3. Ensure the emails do have attachments.
4. Ensure that the data attachment is before the event attachment.
5. Ensure that there is no invalid data in the body of the email, or in the attachment.
6. Do not move emails between folders in excel. A Copy is required for the driver to operate to specification.

8. Appendix

8.1. Appendix A: Gmail Setup for Email Drivers using IMAP [New Google Security Policy]

This knowledge base article details the steps required in configuring the Gmail IMAP server with the mentioned Adroit email drivers, in line with Gmail's new security policies implemented as of **30 May 2022**. For further information, refer to the following Google Account Help article: <https://support.google.com/accounts/answer/6010255?hl=en>

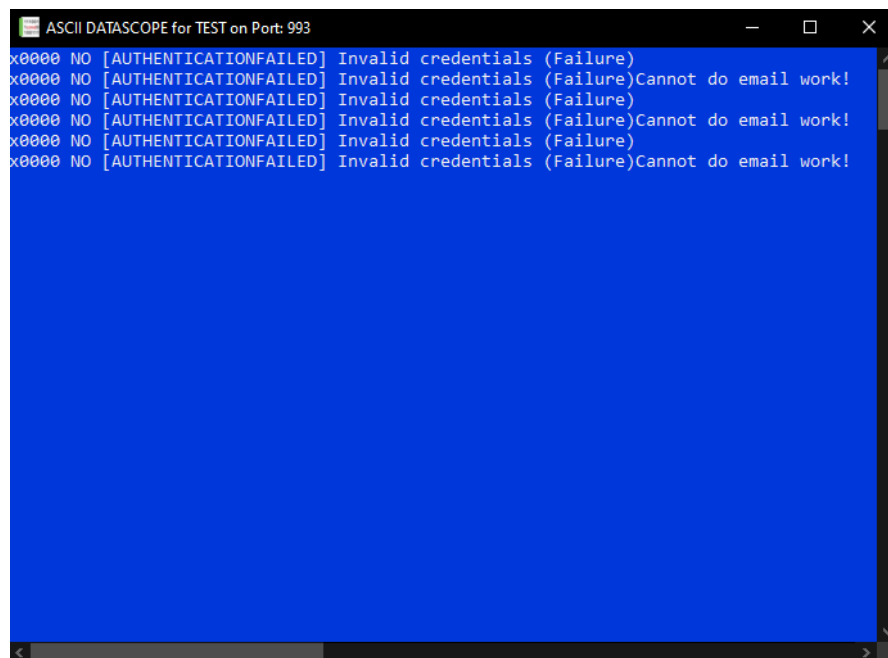
The opening article states the following:

To help keep your account secure, from **May 30, 2022**, Google no longer supports the use of third-party apps or devices which ask you to sign in to your Google Account using only your username and password.

Important: This deadline does not apply to Google Workspace or Google Cloud Identity customers. The enforcement date for these customers will be announced on the Workspace blog at a later date.

For more information, continue to read.

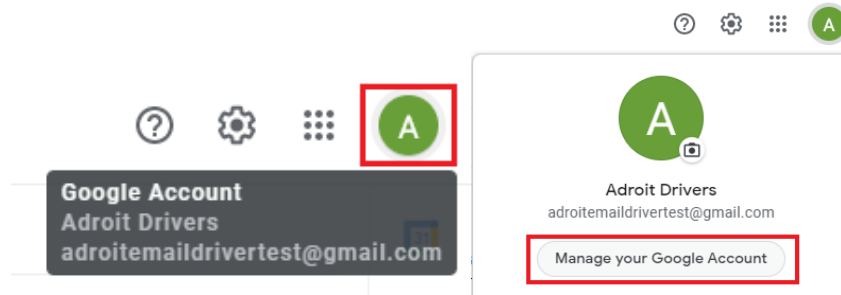
Previously, Adroit email driver's required full access to your Gmail account, and subsequently your Google account, to allow access into your inbox using the Gmail IMAP server. This required adjusting a security setting on your Google account to allow 3rd party applications to access your account. If you have noticed that Adroit has not been able to successfully connect to the Gmail IMAP server, typically stating an authentication error, the problem could be related to this issue.



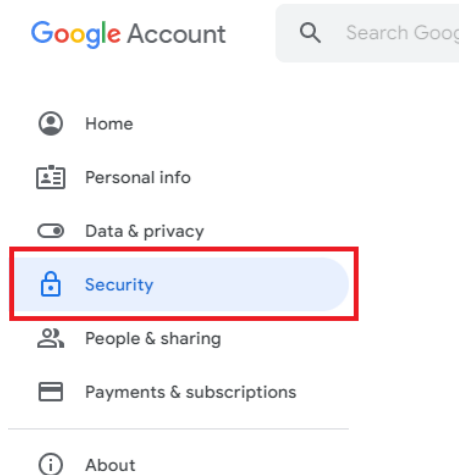
Example of MYCITY Email Client Protocol Driver device authentication error [x0000 NO [AUTHENTICATIONFAILED] Invalid Credentials (Failure) Cannot do email work!]

To access the setting in question, follow the steps provided:

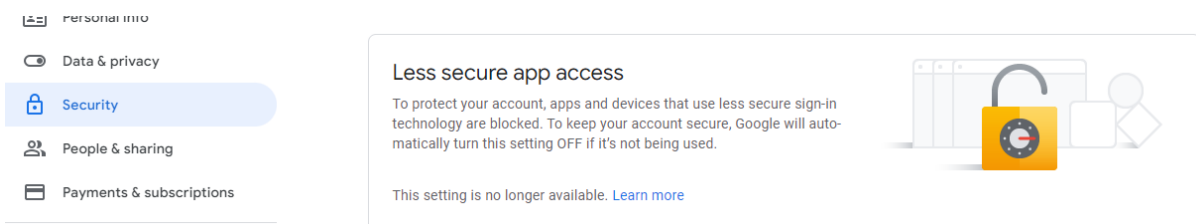
- 1) Accessing from Gmail directly:
 - a. **Log in** to your Gmail account
 - b. Click on the Google Account indicator in the top right-hand corner of the screen. Click on **Manage your Google Account**



c. On the page that opens, click on the **Security** tab located on the left-hand side



d. Under the Security section, there used to be a section **Less secure app access** which was required to be **On**. However, this setting has since been removed

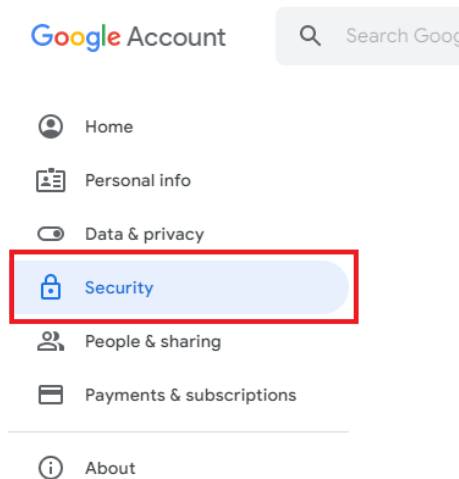


Please Note: if you are unable to see this setting, you probably have **2-Step Verification** turned on. In the past, Adroit would require this be turned off, but please read further as this is now required to be on.

- 2) Accessing account security directly
 - a. **Log in** to your Google Account
 - b. Access your **Google Account**: <https://myaccount.google.com/>
 - c. Follow from Step 1)c outlined previously

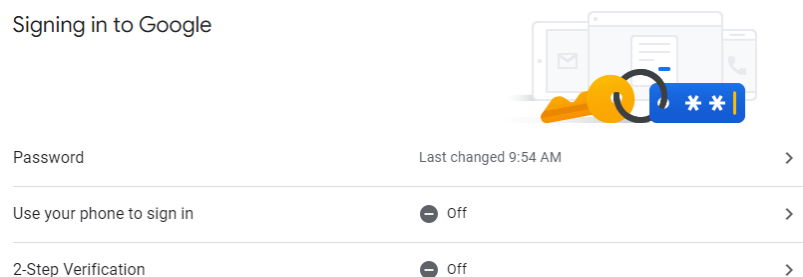
To fix this issue, the user will need to create an App Password that Adroit will then use for authentication with Gmail. This requires that 2-Step Verification is enabled. The steps provided below are summarised and can also be read from here, should you require additional information: <https://support.google.com/accounts/answer/185833?hl=en#:~:text=Create%20%26%20use%20App%20Passwords>

- 1) **Log in** to your Google/Gmail Account
- 2) Access your **Google Account**: <https://myaccount.google.com/>
- 3) Click on the **Security** tab on the left-hand side of the screen



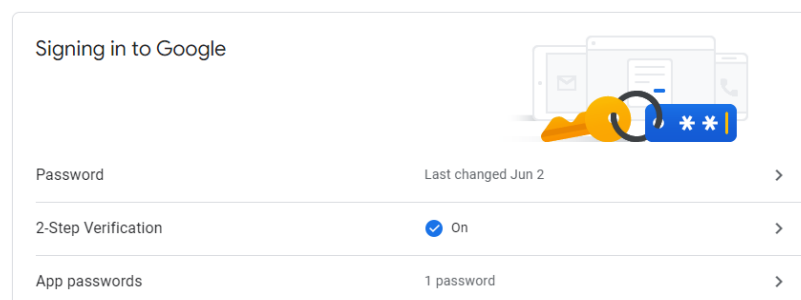
4) Navigate the menu to the section **Signing in to Google**

Signing in to Google



5) Enable **2-Step Verification** if it is not already. If has not been enabled, do so and follow the prompts by Google. You will need a cellphone number to continue.

6) After setting up 2-Step Verification, click on the new option under the **Signing into Google** called **App passwords**



7) After verifying it is you, you will see a list of App passwords currently registered. Generate a new App password by clicking the **Select app** dropdown box, and selecting the **Mail** option, then the **Select device** dropdown box, selecting the **Windows Computer** option. Click on the **Generate** button.

Select the app and device you want to generate the app password for.

Mail

Select device

iPhone

iPad

BlackBerry

Mac

Windows Phone

Windows Computer

Other (Custom name)

GENERATE

- 8) A message will appear with a 16-character randomly generated password. Copy this and save it somewhere secure. This is the password you will use for Adroit to access the Gmail account.

Generated app password

Your app password for Windows Computer

•llg a/fi wakj mojs

How to use it

1. Open the "Mail" app.
2. Open the "Settings" menu.
3. Select "Accounts" and then select your Google Account.
4. Replace your password with the 16-character password shown above.

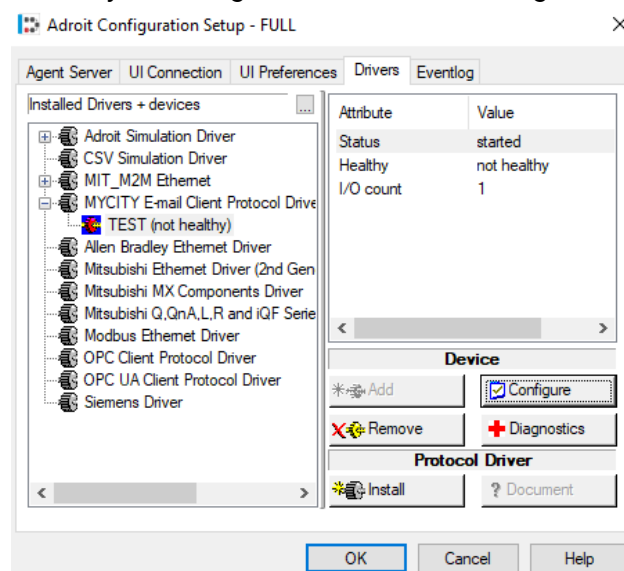
Just like your normal password, this app password grants complete access to your Google Account. You won't need to remember it, so don't write it down or share it with anyone.

[Learn more](#)

DONE

- 9) Open the **Classic Adroit Setup** and navigate to the device experiencing the issue.

- 10) Open the device configuration by selecting the device and clicking **Configure**.



- 11) Ensure the details of the device have been input correctly and place the App password previously generated into the **Login Password/Password** input field.

MYCITY E-mail Client Protocol Driver : TEST

Port Selection
☐ Enable Secondary
☒ Primary
☐ Secondary

Local Device Settings
Station Id

OK
Cancel

Common Mail Settings (*Resource key)
Resource

Mailbox Type
☒ IMAP
☐ POP

Mail Server
Mail Port

Login Name
Login Password

Mailbox Poll Period (ms)
☒ Move Used/ToDo Messages to Backup/ToDo Mailboxes.
☐ Delete Used/ToDo Messages.
☐ Delete Spam Messages.
☒ Server uses SSL.
☒ Detailed Debugging on Datascope.

Incoming Mailbox
Backup Mailbox
ToDo Mailbox
Error Mailbox

12) Exit the Device configuration by clicking on **OK**. You may be prompted to change the current resource for all mailboxes if you are using the MyCity Email driver, for example. You can create a new resource or override it for all devices using the same resource.

13) The device should now be marked as healthy. Open the datascope to see if the error is persisting.

Please Note: you may need to **restart** your **Agent Server** before the changes are applied.

Please Note: The App password generated to allow access to Gmail's Email Server does not rely on the verification that 2-Step Verification provides, as it allows the third-party application limited control direct access to your Gmail account. Please keep this password private and secure.

However, considering these changes, Google may continue to further change and evolve their security policies, with or without notice, which can affect the ability of the Adroit email drivers to function as expected. It is therefore recommended that users provide an alternative email with a different provider as a backup solution, or to migrate to another email provider.

Provided is a list of tested email providers that can be utilized with Adroit email drivers as an alternative to Gmail, and more will be added in the future:

- Outlook
-

Please contact Adroit Support should you wish to enquire about compatibility with other email service providers.

For further details on device configuration, please refer to the respective Adroit driver documentation. For any additional information or queries, please feel free to contact [Adroit Support](#) or search our [Knowledge Base](#).